



Customer Shopping Behavior Analysis

Uncovering insights from **3,900 transactions** across demographics, product preferences, and subscription behavior to guide strategic business decisions.

Dataset at a Glance

3,900

Total Transactions

Across all product categories

18

Data Columns

Demographics, purchases & behavior

37

Missing Values

In Review Rating column only

50

Unique Locations

Customer geographic spread

Key features include customer demographics (age, gender, location, subscription), purchase details (item, category, amount, season), and shopping behavior (discounts, frequency, review ratings, shipping type).



Data Preparation in Python

→ Loading & Exploration

Imported dataset via `pandas`; used `df.info()` and `.describe()` for structure and summary statistics.

→ Cleaning

Imputed missing **Review Rating** values using median rating per product category; standardized columns to `snake_case`.

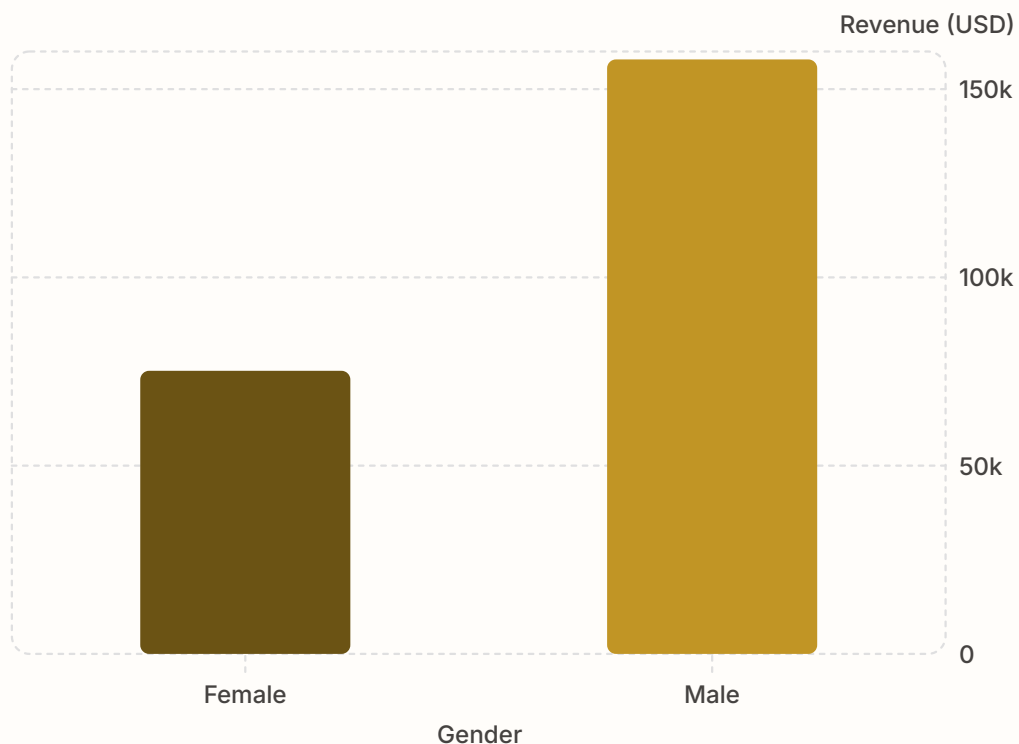
→ Feature Engineering

Created `age_group` bins and `purchase_frequency_days`; dropped redundant `promo_code_used` column.

→ Database Integration

Loaded cleaned DataFrame into **MySQL** for SQL-based analysis.

Revenue by Gender & Subscription Status



Male Customers Drive 2× Revenue

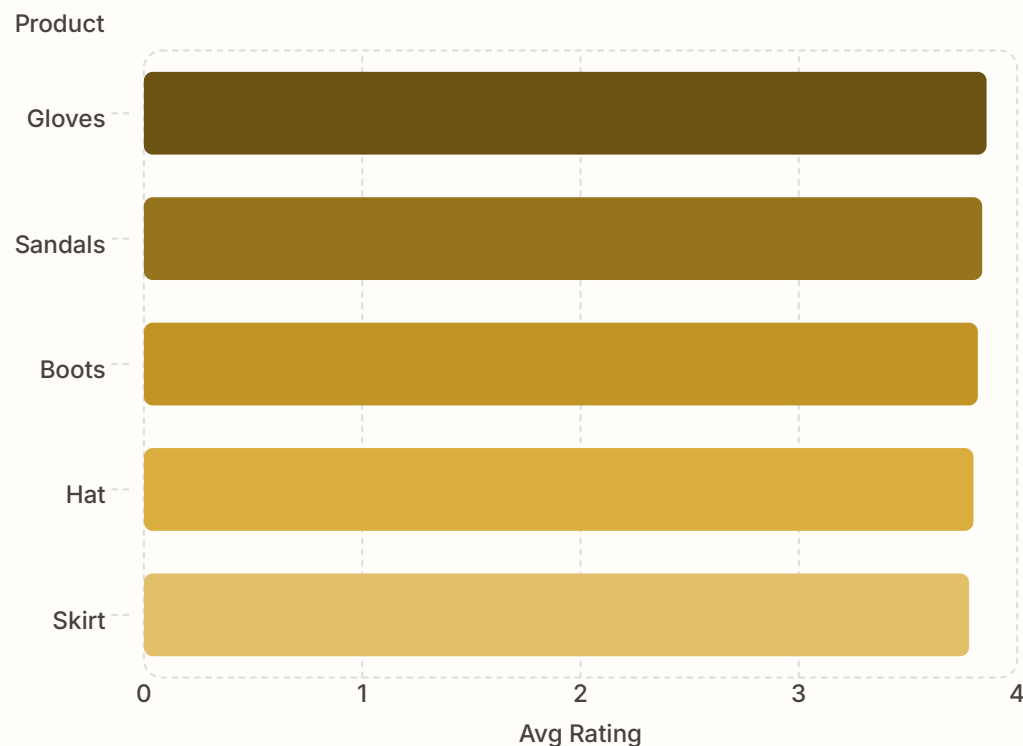
Male customers generated **\$157,890** vs. **\$75,191** for females — a significant gap worth investigating for targeted marketing.

Subscription Revenue Split

Non-subscribers (73% of customers) generated **\$170,436** in total revenue, while subscribers (27%) contributed **\$62,645**. Average spend is nearly identical (~\$59.50), suggesting subscription benefits may not yet be compelling enough to convert high spenders.

Top Products & Discount Behavior

Top 5 Products by Review Rating



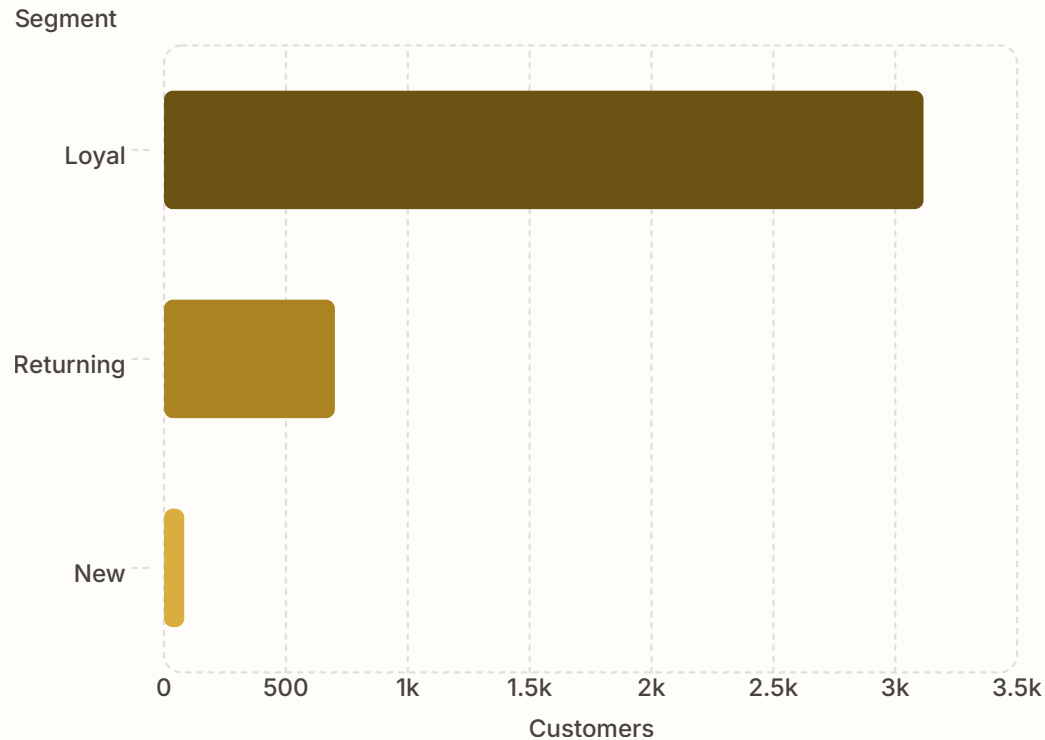
Most Discount-Dependent Products

These 5 products have the highest percentage of discounted purchases:

- **Hat** — 50.00%
- **Sneakers** — 49.66%
- **Coat** — 49.07%
- **Sweater** — 48.17%
- **Pants** — 47.37%

📌 Nearly half of sales for these items rely on discounts — a key margin consideration.

Customer Segmentation



Segment Breakdown

Customers classified by purchase history into three segments:

Loyal — 3,116 customers

The dominant segment, representing the core revenue base.

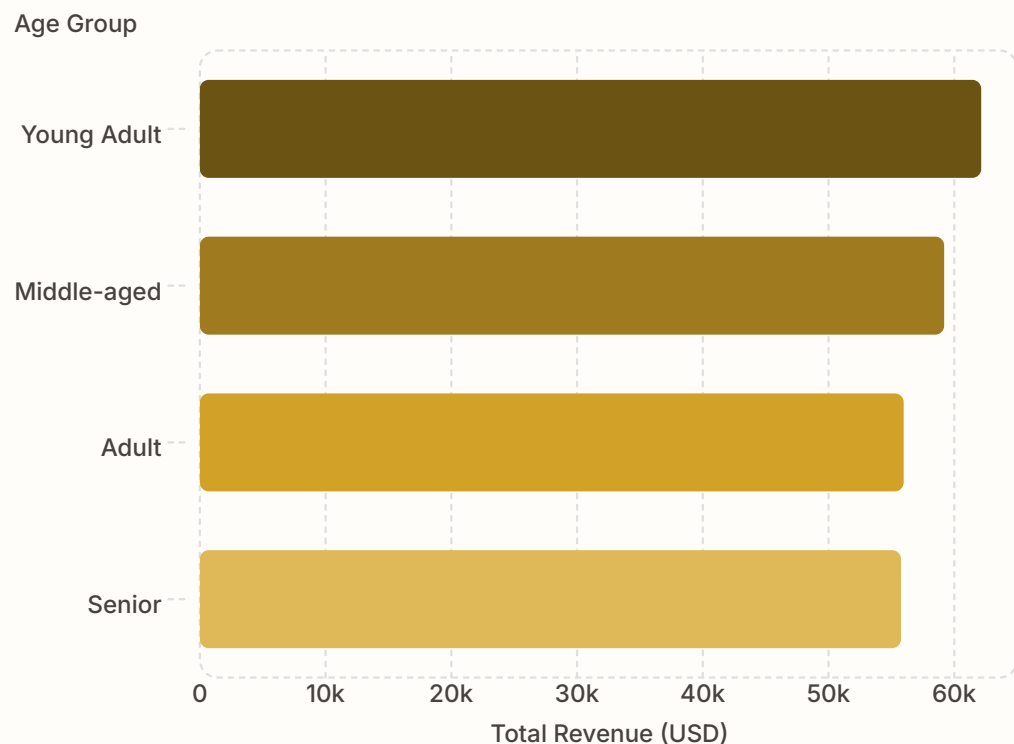
Returning — 701 customers

Active but not yet fully loyal; prime candidates for loyalty programs.

New — 83 customers

Smallest segment; opportunity for onboarding and retention campaigns.

Revenue by Age Group



Young Adults Lead Revenue

Young Adults contribute the highest revenue at **\$62,143**, followed closely by Middle-aged (**\$59,197**). Revenue is relatively evenly distributed across all four age groups, with only a **\$6,380** difference between the highest and lowest.

i Customer age ranges from 18 to 70, with a mean age of 44. The age_group column was created by binning customer ages for this analysis.

Top Products by Category

Most purchased items within each of the four product categories:

Accessories

1. Jewelry — 171 orders
2. Sunglasses — 161 orders
3. Belt — 161 orders

Clothing

1. Blouse — 171 orders
2. Pants — 171 orders
3. Shirt — 169 orders

Footwear

1. Sandals — 160 orders
2. Shoes — 150 orders
3. Sneakers — 145 orders

Outerwear

1. Jacket — 163 orders
2. Coat — 161 orders

Power BI Dashboard

An interactive **Customer Behavior Dashboard** was built to present all insights visually, with filters for subscription status, gender, category, and shipping type.



3.9K Customers

Total transaction count



\$59.76 Avg Purchase

Mean spend per order



3.75 Avg Rating

Overall review score



27% Subscribers

Donut chart: Yes 27% / No 73%



Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits to convert high-spending non-subscribers.



Loyalty Programs

Reward repeat buyers to advance them into the "Loyal" segment (currently 3,116 customers).



Review Discount Policy

Balance sales boosts with margin control — nearly 50% of top discount-dependent products rely on promotions.



Targeted Marketing

Focus on high-revenue age groups (Young Adults) and express-shipping users for maximum ROI.